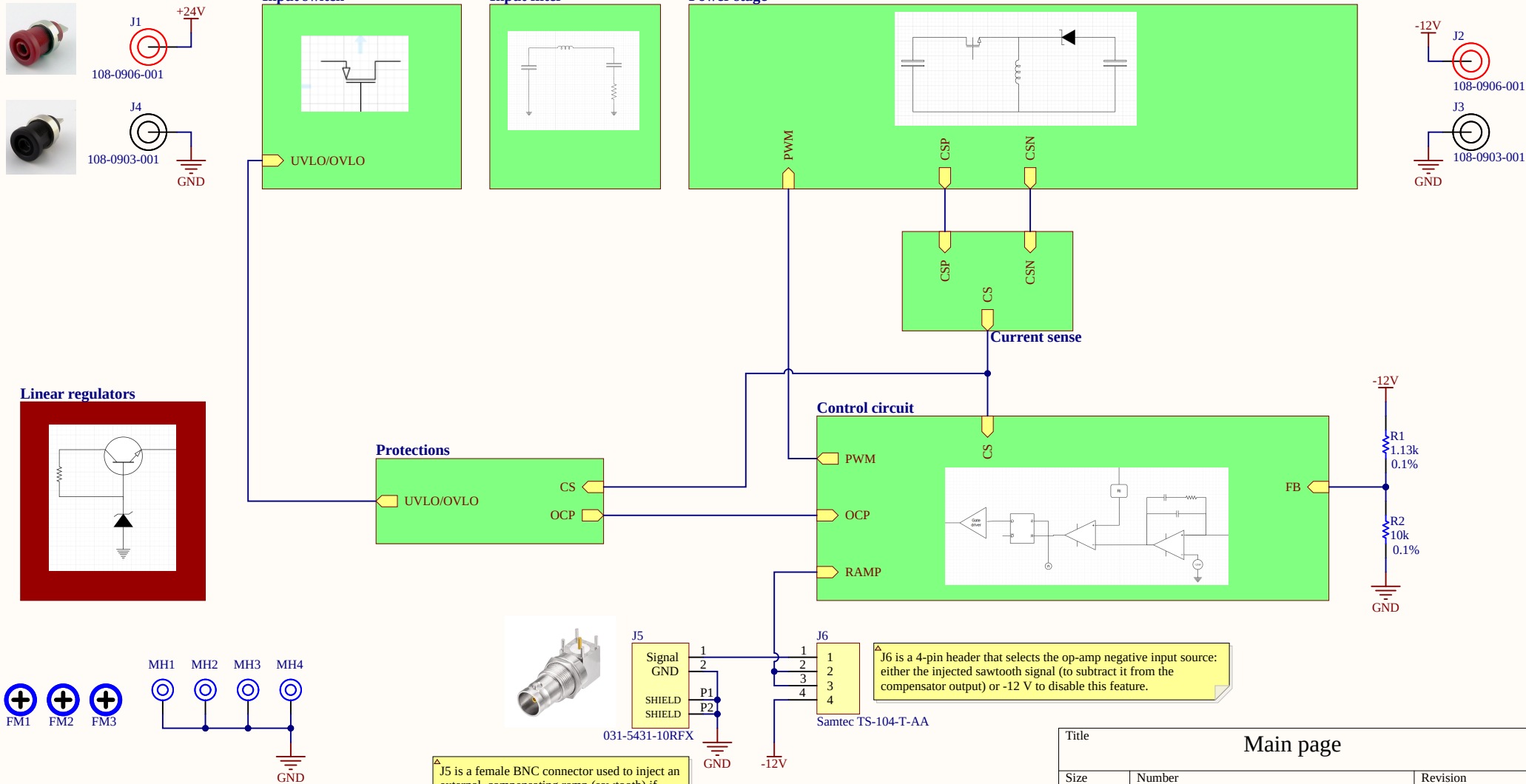


Inverting Buck-Boost



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Power stage

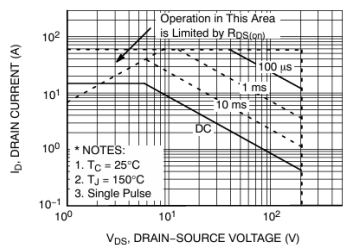
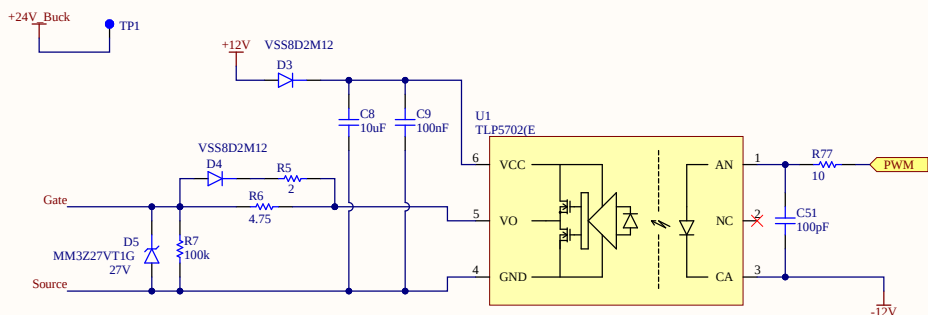
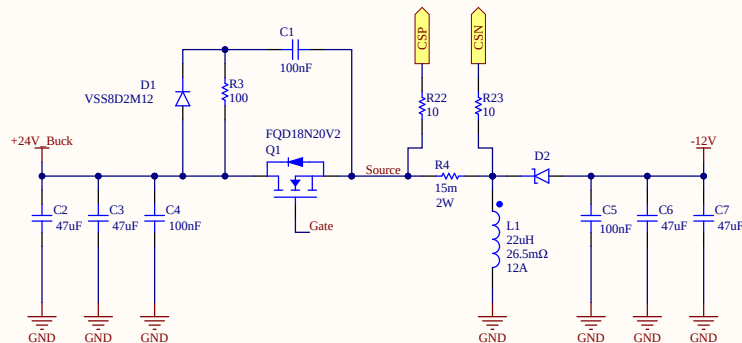


Figure 9. Maximum Safe Operating Area

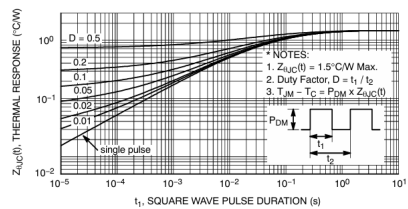


Figure 11. Transient Thermal Response Curve

Latch-off / hiccup OCP:
Q1 hits 11A once, OCP trips, gate goes off, converter stops (or waits to retry). The 396W linear event happens once → single pulse is correct.
Cycle-by-cycle OCP:
Q1 hits 11A, gets turned off for that cycle, then turns back on next cycle and hits 11A again... repeating every switching period for several ms during a sustained overload. Now the high-power linear crossing repeats every cycle → this is a pulse train, and you should use the duty-cycle curve, NOT single pulse.

FQD18N20V2 (Q1) — SOA CALCULATION

Package: DPAK | 200V | 15A | $R_{th}(j-c) = 1.5^{\circ}C/W$ | $T_{jmax} = 150^{\circ}C$
 $T_A = 40^{\circ}C$ (max ambient)
SOA points (Fig.9 @ $V_{DS} = 36V$, $25^{\circ}C$):
 $t_1 = 1ms \rightarrow ISOA = 20A$
 $t_2 = 10ms \rightarrow ISOA = 7A$
POWER-LAW FIT:
 $SOA(t) = a \times t^m$ (t in seconds)
 $m = \ln(20/7) / \ln(1ms/10ms) = 1.050 / 2.303 = -0.456$
 $a = 20(1ms)^{(-0.456)} = 0.8575$
 $SOA(t) = 0.8575 \times t^{(-0.456)}$

CASE 1 — NORMAL SWITCHING

- ON-state V_{DS} in RDS(on) region, SOA-safe (low V_{DS})
- Switching edges: cross active region but last ~20ms < 10μs → far inside SOA pulse limit
- $ID_{peak} = 5.31A < SOA(10μs) \rightarrow$ far inside SOA ✓

CASE 2 — STARTUP (OCP clamp = 11A)

$V_{DS} = V_{in} + |V_{out}| = 36V$
 $P_{pulse} = V_{DS} \times I_{OCP} = 36 \times 11 = 396W$
 $Z_{th(j-c)}$ read from Fig.11 SINGLE-PULSE curve:
 $t = 1ms \rightarrow Z_{th} = 0.17$ $t = 2ms \rightarrow Z_{th} = 0.25$ $t = 3ms \rightarrow Z_{th} = 0.30$
(arbitrary sweep points to locate the current/pulse limit)

$T_j = T_A + Z_{th}(t) \times P_{pulse}$
 $ISOA_{derated} = SOA(t) \times (T_{jmax} - T_j) / (T_{jmax} - 25)$

$t = 0.64ms$: $T_j = 40 + 0.136 \times 396 = 94^{\circ}C$
 $SOA_{25} = 0.8575 \times (0.64ms)^{-0.456} = 24.5A$
 $der = 24.5 \times (150 - 94) / 125 = 11.0A = 11A \blacktriangleleft$ LIMIT
 $t = 1ms$: $T_j = 40 + 0.17 \times 396 = 108^{\circ}C$
 $SOA_{25} = 20.0A$, $der = 20.0 \times (150 - 108) / 125 = 6.8A < 11A \times$

CHECK: $I_{OCP} = 11A$ vs $ISOA_{derated}$
 $t \leq 0.64ms$: $11A \leq ISOA_{derated}$ ✓ PASS
 $t > 0.64ms$: $11A > ISOA_{derated}$ ✗ FAIL

RESULT

Max safe linear pulse at 11A/36V ($T_A = 40^{\circ}C$) = 0.64ms
→ OCP must turn Q1 OFF within 0.64ms of reaching 11A.
Cycle-by-cycle: one period = 3.3μs @ 300kHz, far inside the 0.64ms safe window ✓

ANALYSIS

$V_{in} = 24V$ | $V_{out} = -12V$ | $I_{out} = 3A$ | $f_s = 300kHz$ | $D = 0.333$ | $eff = 88\%$

INDUCTOR — SRP1245A-220M (22μH)

$IL_{avg} = I_{in} + I_{out} = 1.705 + 3 = 4.705A$
 $\Delta IL = V_{in} \times D / (f_s \times L) = 1.21A$
 $IL_{peak} = IL_{avg} + \Delta IL / 2 = 5.31A$
 $IL_{ripple_ratio} = 1.21 / 4.705 = 25.7\%$
 $L_{crit} = (D)^2 \times R / 2f_s = 2.97μH$
CCM confirmed: $L = 22μH \gg L_{crit} = 2.97μH$ ✓

MOSFET — FQD18N20V2 (200 V, 15 A, 140 mΩ)

$V_{DS_stress} = V_{in} + V_{out} = 24 + 12 = 36V$
 $V_{DS_rated} = 36 \times 1.2 = 43.2V \rightarrow 600V$ ✓ (20% derating)
 $ID_{rated} \geq IL_{peak} = 5.31A \rightarrow 15A$ ✓
 $ID_{rms} = IL_{avg} \times \sqrt{D} = 2.72A$
 $P_{cond} = ID_{rms}^2 \times R_{DS(on)} \times 1.3 = 2.72^2 \times 0.637 = 4.71W$ (30% derating)
 $P_{sw} = 0.5 \times V_{DS} \times IL_{peak} \times (t_r + t_f) \times f_s = 0.5 \times 36 \times 5.31 \times 16n \times 300k = 0.46W$
 $P_{total} = 4.71 + 0.57 = 5.12W$

DIODE — STPS10L60S (60V, 10A, $V_F = 0.4V$, DPAK)

$V_r_stress = V_{in} + V_{out} = 36V$
 $V_r_rated = 36 \times 1.2 = 43.2V \rightarrow 60V$ ✓
 $ID_{avg} = I_{out} = 3A$
 $ID_{peak} = IL_{peak} = 5.31A$
 $P_{diode} = V_F \times I_{out} \times (1-D) = 0.4 \times 3 \times 0.667 = 0.80W$

OUTPUT CAPACITOR — KRM55WR71H476MH13K × 2 (47μF, 50V, X7R)

$C_{min} = I_{out} \times D / (f_s \times \Delta V_{out}) = 27.7μF$
 $C_{nominal} = 2 \times 47 = 94μF$
 $C_{derated} @ 12V = 2 \times 39.6 = 79.2μF > 27.7μF$ ✓
 $\Delta V_{out_target} = 1\% \times 12 = 120mV$
 $ESR_{max} = 0.5 \times \Delta V_{out} / \Delta IL = 49.6mΩ$
(the ESR can contribute max by a 50% to the total allowed ripple)
 $ESR_{eff} = 4mΩ / 2 = 2mΩ \ll 49.6mΩ$ ✓
 $IC_{rms} = I_{out} \times \sqrt{(D/D')} = 2.12A$
 $IC_{rms/cap} = 2.12 / 2 = 1.06A$ ✓

INPUT CAPACITOR — KRM55WR71H476MH13K × 2 (47μF, 50V, X7R)

$C_{min} = I_{out} \times D / (f_s \times \Delta V_{in} \times D') = 20.8μF$
 $C_{nominal} = 2 \times 47 = 94μF$
 $C_{derated} @ 24V = 2 \times 26.6μF = 53.2μF > 20.8μF$ ✓
 $\Delta V_{in_target} = 1\% \times 24 = 240mV$
 $ESR_{max} = 0.5 \times \Delta V_{in} / IL_{peak} = 22.6mΩ$
 $IC_{in_rms} = IL \times \sqrt{(D \times D')} = 2.22A$ ✓
 $V_{rated} = 32 \times 1.5 = 48V \rightarrow 50V$ ✓
 $C_{in} \text{ impedance } Z_{cin} = 1 / (2\pi \times f_s \times C_{in}) + ESR_{cin} = 0.413Ω$
 $C_{damp} \text{ impedance } Z_{cdamp} = 1 / (2\pi \times f_s \times C_{damp}) + ESR_{damp} = 0.013Ω$
 $IRMS \text{ Total Input Caps } (C_{in} + C_{damp}) = IL \times \sqrt{(D \times D')} = 2.22A$
 $IRMS \text{ C}_{damp} = Z_{cdamp} / (Z_{cdamp} + Z_{cin}) = 2.16A$ ✓

POWER BUDGET

$P_{mosfet} = 5.17W$
 $P_{diode} = 0.80W$
 $P_{total} = 6.08W$
 $\eta_{estimated} = P_{out} / (P_{out} + P_{loss}) = 36 / (36 + 6.08) = 85.6\%$

MOSFET: FQD18N20V2

Driver: TLP5702(E)

$V_{DRV} = 24V$

Method 1 — Total Gate Charge

$Q_g = 20nC$

$I_{drv,max} = 2.5A$

$t_r = Q_g / I_{drv,max} = 8ns$

$t_f = Q_g / I_{drv,max} = 8ns$

Method 2 — Miller Charge :

$Q_{gd} \approx 7nC$

$V_{plateau} \approx 7.8V$

$I_{g,on} = \min[(V_{DRV} - V_{plateau}) / R_{gon}, I_{drv,max}]$

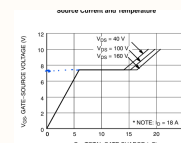
$I_{g,off} = \min[(V_{DRV} - V_{plateau}) / R_{goff}, I_{drv,max}]$

$t_r = Q_{gd} / I_{g,on} \approx 2.8ns$

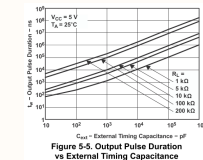
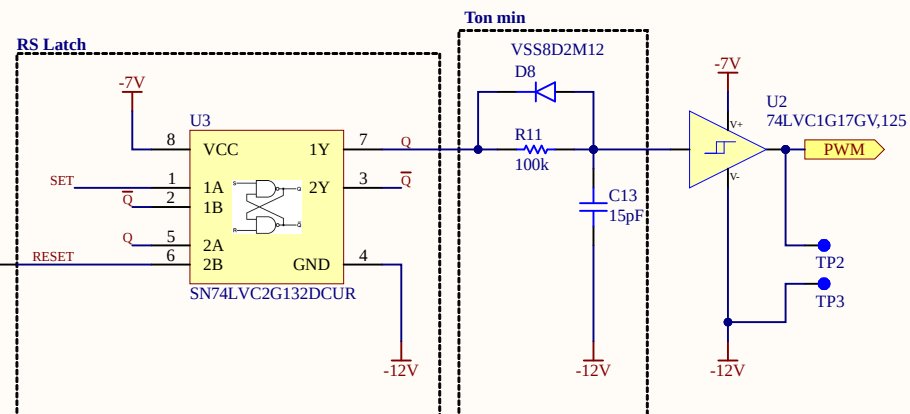
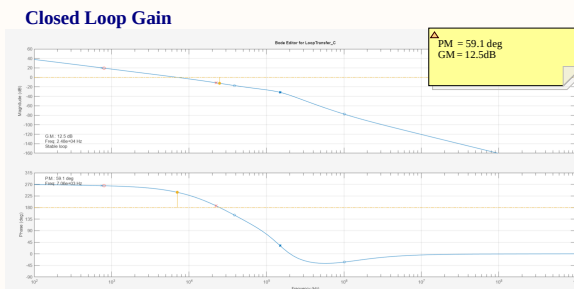
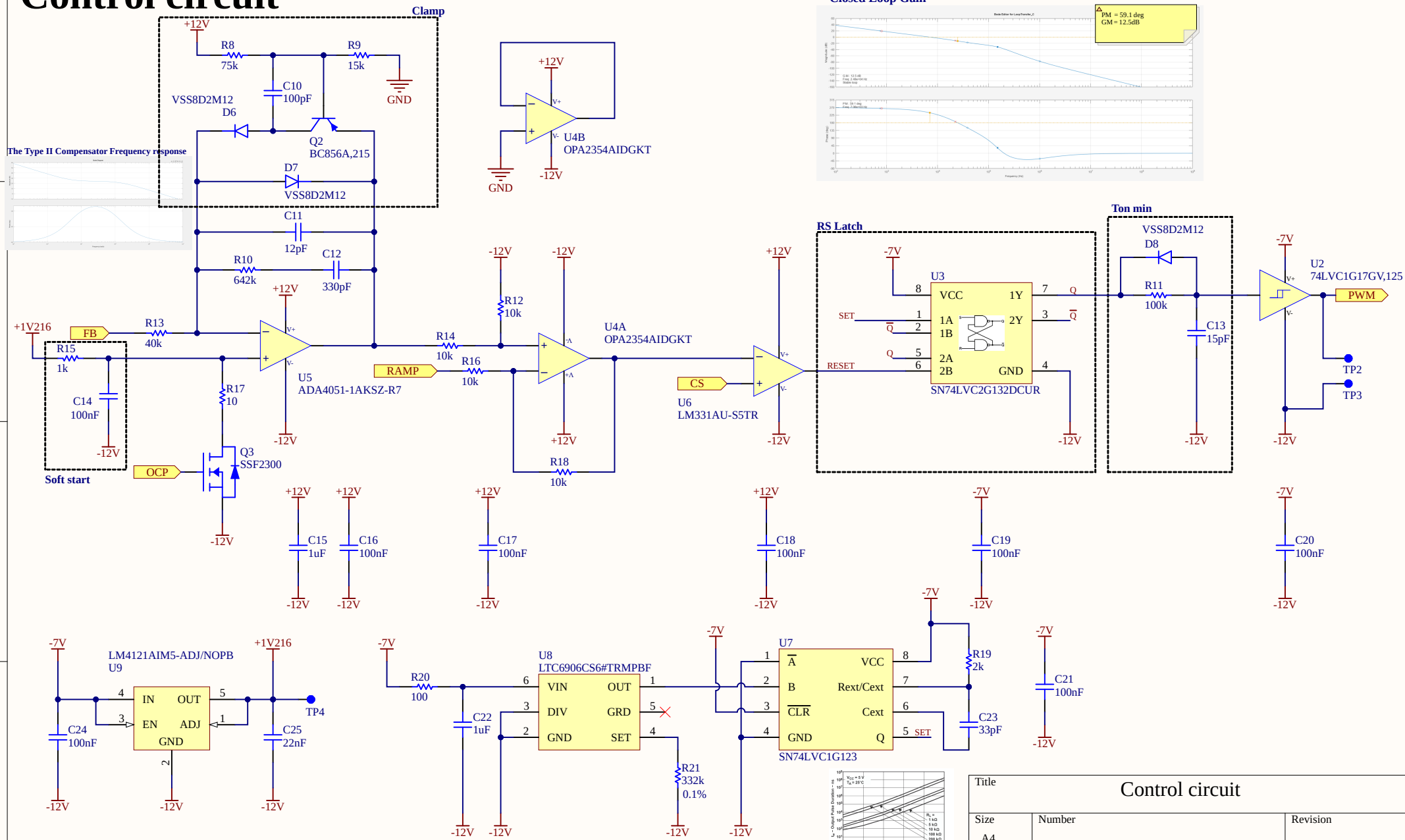
$t_f = Q_{gd} / I_{g,off} \approx 2.8ns$

Switching Loss:

$P_{sw} = 0.5 \times V_{DS} \times ID \times (t_r + t_f) \times f_s$

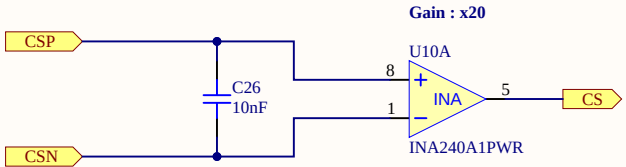


Control circuit

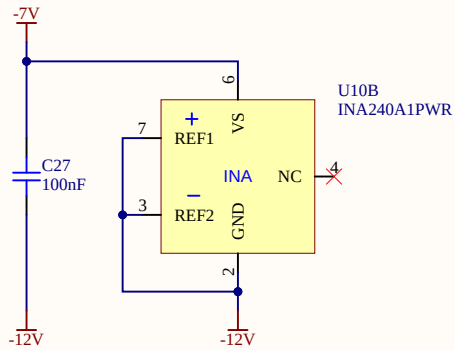


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Current sense



^A Total current gain = 0.015 x 20 = 0.3



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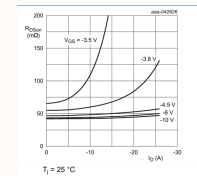


Fig. 7. Drain-source on-state resistance as a function of drain current; typical values

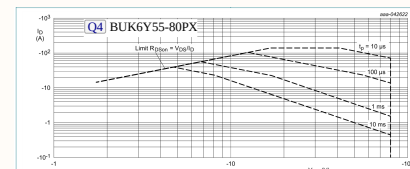
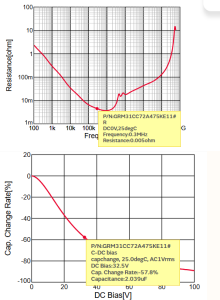
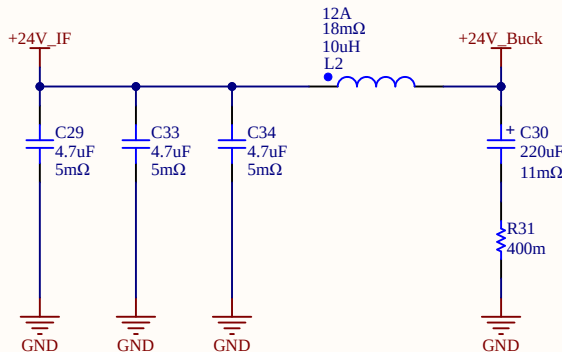
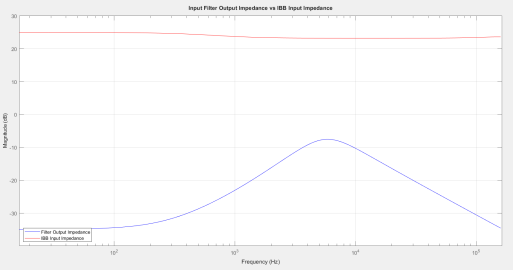


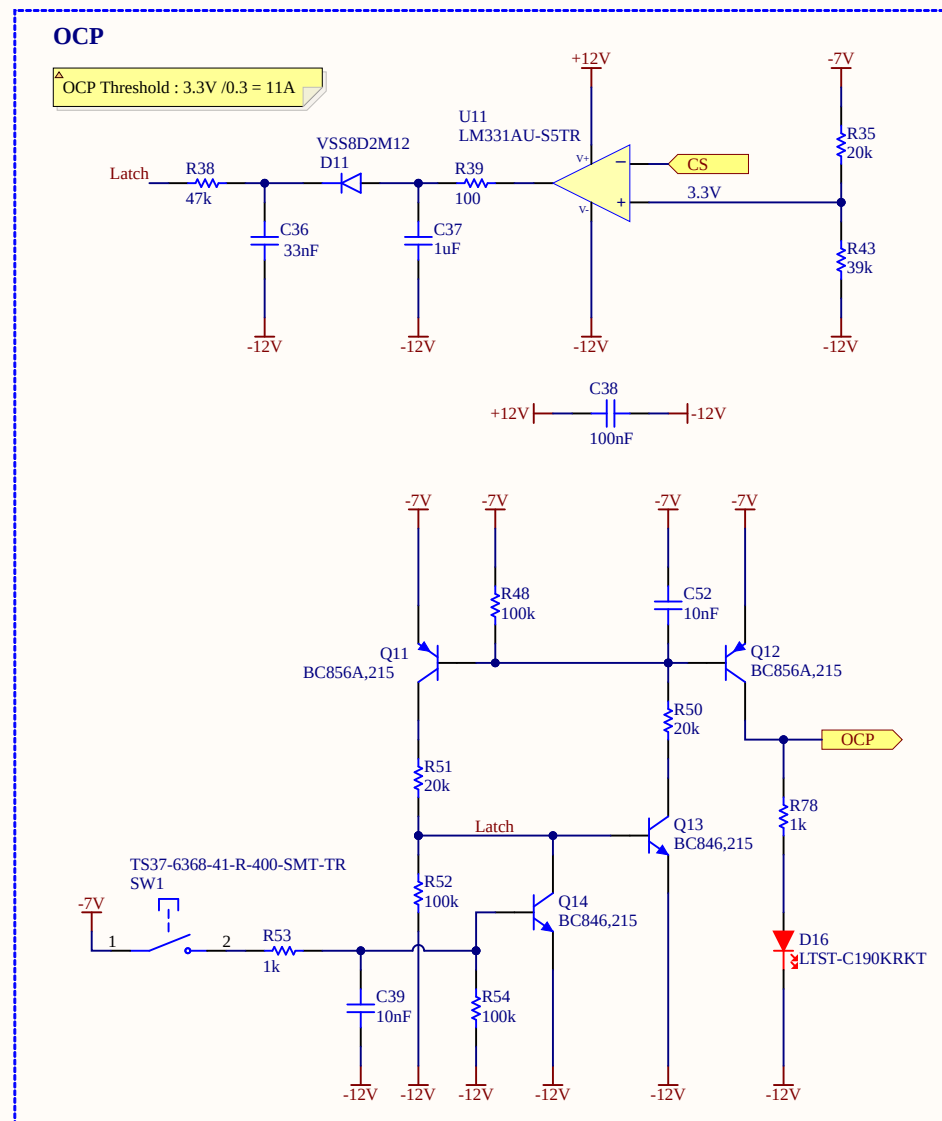
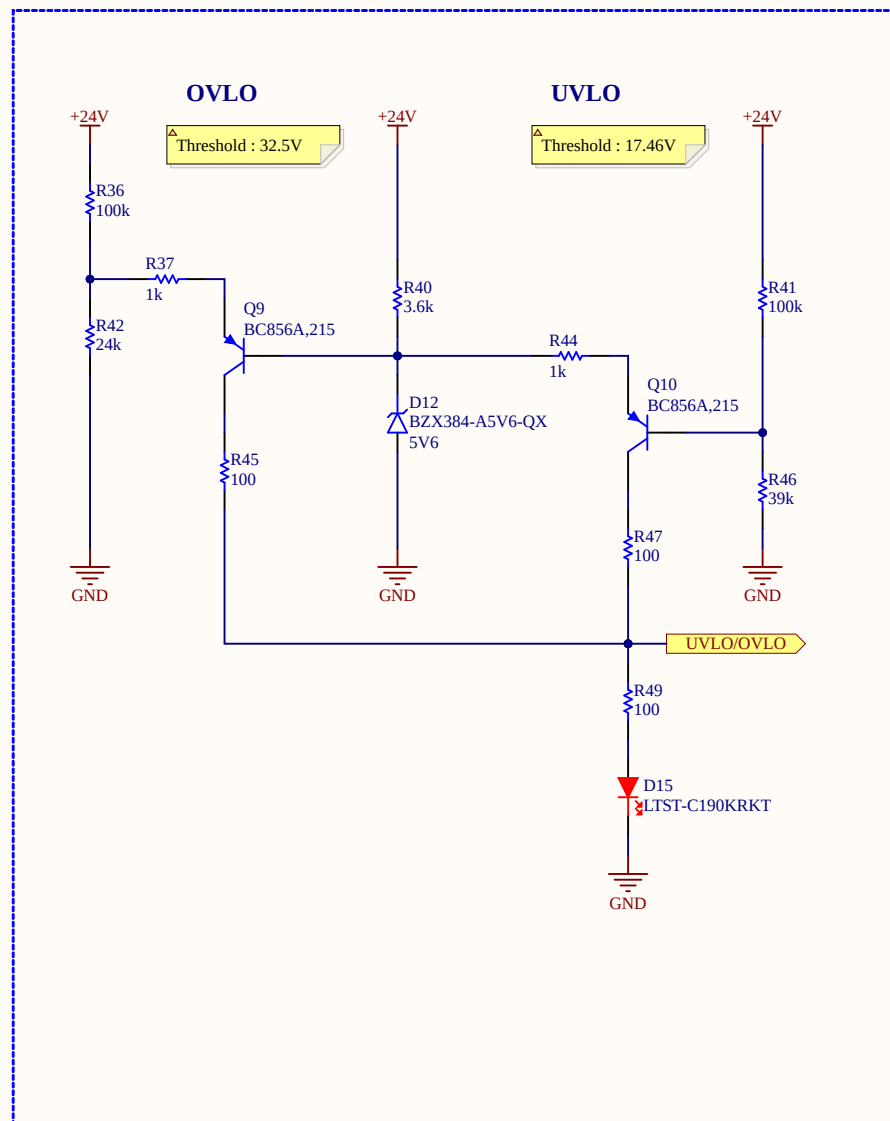
Fig. 3. Safe operating area; junction to mounting base; continuous and peak drain currents as a function of drain-source voltage

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^ INPUT EMI FILTER : Design Notes		
Standard: EN 55032:2015 Class B — Conducted Emissions		
Limit @ 300kHz = 56 dBμV (quasi-peak)		
Frequency range: 150kHz – 30MHz		
SPECIFICATIONS:		
Vin = 24V Vout = -12V Iout = 3A fs = 300kHz D = 0.333 Cin = 53.2uF (derated @24V)		
STEP 1 : NOISE LEVEL ESTIMATE		
First harmonic amplitude:		
I1 = (2×IL×sin(π×D)) / π = (2×4.705×sin(π×0.333)) / π = 2.60A		
Noise voltage across Cin at fs:		
Vnoise = I1 / (2π×fs×Cin) = 2.60 / (2π×300k×53.2u) = 25.9mV = 25.9×10³ μV		
Estimated noise level:		
Noise dBμV = 20log(25.9×10³) = 88.3 dBμV		
Required attenuation:		
Att = Noise dBμV - Vmax = 88.3 - 56 = 32.3 dB		
Adding 10dB safety margin:		
Att_target = 32.3 + 10 = 42.3 dB		
STEP 2 : INDUCTOR SELECTION		
Lf = 10uH → Selected: 10uH, 12A, SMD		
STEP 3 : CAPACITOR SELECTION		
Resonance one decade below fs:		
Cf = Cin / (Cin×Lf×(2π×fs/10)² - 1) = 53.2u / (53.2u×10u×(2π×30k)² - 1) = 3μF		
Required attenuation:		
Cf= 1/Lf × (10^(Att/40)/(2π×fs))² = 5μF		
Select higher value: Cf = max(5μF,3μF) = 5μF→ 3x 4.7μF selected derated to 2uF @Vin,max = 32.5V		
NOTE: Cf is a SEPARATE dedicated filter capacitor distinct from Cin.		
fc_filter = 1/(2π×√(Lf×Cf)) = 1/(2π×√(10u×6μF)) = 20.5kHz		
Attenuation verification:		
A = 40×log(fs/fc_filter) = 40×log(300k/20.5k) = 46.6dB >> 42.3dB ✓		
STEP 4 : DAMPING CAPACITOR		
Cd ≥ 4×Cin = 4×53.2u = 212.8uF → 220uF selected		
Required ESRd:		
ESRd ≤ √(Lf/Cin) = √(10u/53.2u) = 0.434Ω		
Damping resistor power:		
P_Rdamp ≈ 0.5×Δlin²×Rdamp = 0.5×(1.21)²×1.5 = 0.8W		
Selected Cd: 220uF 35V polymer electrolytic ESR_polymer ≈ 11mΩ < 434mΩ ✓		
NOTE: Polymer ESR too low, added series resistor Rd = 0.4Ω in series with Cd to meet ESRd target		
Voltage rating: Vin_max×1.25 = 32×1.25 = 40V → 50V ✓		
Cin impedance Zcin = 1/(2π×fs×Cin) + ESRcin = 0.413Ω		
Cdamp impedance Zcdamp = 1/(2π×fs×Cdamp) + ESRdamp = 0.013Ω		
IRMS Total Input Caps (Cin +Cdamp) = IL×√(D×D') = 2.22A		
IRMS Cdamp (current divider) = Zcin / (Zcdamp +Zcin) = 0.062A✓		
STEP 4 : MIDDLEBROOK CRITERION		
Zin_conv = Vin²×η/Pout = 576×0.88/36 = 14.1Ω		
Full frequency spectrum verified using matlab: see Plot		
EMC STANDARD : EN 55032:2015		
EN 55032 "Electromagnetic Compatibility of Multimedia Equipment" . It defines conducted emission limits on the power port for Class B equipment (residential environment):		
150kHz – 500kHz : 56–66 dBμV (quasi-peak)		
46–56 dBμV (average)		
500kHz – 30MHz : 56 dBμV (quasi-peak)		
46 dBμV (average)		
Worst case for this design at fs=300kHz: 56 dBμV QP.		
Reference : Conducted EMI Characteristics And Mitigation Technique		
https://www.ti.com/lit/an/snva489c/snva489c.pdf		

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Protections



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2SC6076 POWER TRANSISTORS — SOA & THERMAL VERIFICATION

DATASHEET (Toshiba 2SC6076):
VCEO = 80V | VCB0 = 160V | IC = 3A DC / 5A PULS Pc = 10W @ Tc=25°C | TJ,max = 150°C
VCE(sat) = 0.5V max @ IC=1A | hFE = 100-450
Thermal resistances (large copper):
Rth(c-a) = 21.2 °C/W
Rth(j-c) = 2 °C/W
(NOTE: Rth values borrowed from ROHM AN 64AN140E (TO-252,JEDEC 2s2p): θJA=23.2, ΨJT=2°C/W → Rth(j-c)=2, Rth(c-a)=21.2. 2SC6076 datasheet gives no Rth; same-package values used as stand-in. Assumes large copper pour — less copper → higher TJ.)
Linear-regulator operation = CONTINUOUS → use DC SOA.

RAIL 1 — +12V REGULATOR (Q15 // Q16)

OPERATING POINT (per device, current shared):
Vce = Vin - Vout = 24 - 12 = 12V
Ic_total = Ilim = 500mA
Ic_perdevice = 500mA / 2 = 250mA

POWER DISSIPATION (per device):
Pd = Vce × Ic_perdevice = 12 × 0.25 = 3.0W (< Pc=10W @ Tc=25°C ✓)
SOA CHECK (DC operation, continuous):
Operating point: Vce=12V, Ic=250mA
DC SOA curve @ Vce=12V :
Ic_SOA(DC, 12V, 25°C) ≈ 0.8A per device
THERMAL:
Rth(c-a) = 21.2 °C/W | Rth(j-c) = 2 °C/W
Tc = TA + Rth(c-a) × Pd = 40 + 21.2 × 3.0 = 103.6 °C
Tj = Tc + Rth(j-c) × Pd = 103.6 + 2 × 3.0 = 109.6 °C ✓ < 150°C

Derated SOA to Tc:
Ic_SOA(DC, Tc) = 0.8 × (150-Tc)/(150-25)
= 0.8 × (150-103.6)/(150-25)
= 0.297A
CHECK: Ic_perdevice = 0.25A < IC_SOA_derated = 0.297A ✓
SOA margin = 0.297/0.25 = 1.19×

RAIL 2 — -7V REGULATOR (Q20 // Q21) — 2 PARALLEL BJTs

OPERATING POINT (per device, current shared 2-way):
Vce = Vin - Vout = 12 - (-7) = 19V
Ic_total = Ilim = 318mA
Ic_perdevice = 318mA / 2 = 159mA
POWER DISSIPATION (per device):
Pd = Vce × Ic_perdevice = 19 × 0.159 = 3.021W (< Pc=10W ✓)

SOA CHECK (DC operation, continuous):
Operating point: Vce=19V, Ic=159mA
DC SOA curve @ Vce=19V : Ic_SOA(DC, 19V, 25°C) ≈ 0.3A per device

THERMAL:
Rth(c-a) = 21.2 °C/W | Rth(j-c) = 2 °C/W
Tc = TA + Rth(c-a) × Pd = 40 + 21.2 × 3.021 = 104 °C
Tj = Tc + Rth(j-c) × Pd = 104 + 2 × 3.021 = 110.1 °C ✓ < 150°C

Derated SOA to Tc:
Ic_SOA(DC, Tc) = 0.3 × (150-Tc)/(150-25)
= 0.3 × 46.0125
= 0.110A
CHECK: Ic_perdevice = 0.159A > IC_SOA_derated = 0.110A ✗ FAIL
SOA margin = 0.110/0.159 = 0.69×

CONCLUSION (2 BJTs): FAIL
Thermal passes (Tj=110°C<150°C) but DC SOA at Vce=19V SOLUTIONS:
(1) Add a 3rd parallel BJT → splits current further
(2) Pre-regulator/series dropper to lower Vce (moves left on SOA, away from second breakdown)
(3) Wider-SOA pass transistor
→ Adopt solution (1): add 3rd parallel device.

RAIL 2 — -7V REGULATOR (Q20 // Q21 // Q26) — 3 PARALLEL BJTs

OPERATING POINT (per device, current shared 3-way):
Vce = 19V
Ic_total = Ilim = 318mA
Ic_perdevice = 318mA / 3 = 106mA

POWER DISSIPATION (per device):
Pd = Vce × Ic_perdevice = 19 × 0.106 = 2.014W (< Pc=10W ✓)

SOA CHECK (DC operation, continuous):
Operating point: Vce=19V, Ic=106mA
DC SOA curve @ Vce=19V :
Ic_SOA(DC, 19V, 25°C) ≈ 0.3A per device

THERMAL:
Rth(c-a) = 21.2 °C/W | Rth(j-c) = 2 °C/W
Tc = TA + Rth(c-a) × Pd = 40 + 21.2 × 2.014 = 82.7 °C
Tj = Tc + Rth(j-c) × Pd = 82.7 + 2 × 2.014 = 86.7 °C ✓ < 150°C

Derated SOA to Tc:
Ic_SOA(DC, Tc) = 0.3 × (150-Tc)/(150-25) = 0.3 × 67.3/125 = 0.162A
CHECK: Ic_perdevice = 0.106A < IC_SOA_derated = 0.162A ✓ PASS
SOA margin = 0.162/0.106 = 1.52×

CONCLUSION (3 BJTs): PASS
3 parallel devices bring per-device current to 106mA, Tj=86.7°C, SOA margin 1.52× at Vce=19V. ✓

EMITTER BALLAST — CURRENT SHARING

RE = 1Ω (R55/R56 and R66/R67)
Purpose: forces current sharing between parallel devices, prevents thermal runaway from VBE mismatch.
Voltage drop across RE:
Rail1: VRE = 0.25A × 1 = 250mV
Rail2: VRE = 0.159A × 1 = 159mV
VRE ≥ 5×ΔVBE rule (ΔVBE≈10mV typical mismatch (between the 2 transistors in //)): 5×10mV = 50mV → Rail1(250mV) ✓ | Rail2(159mV) marginal
PRE: Rail1 = 0.25×1 = 250mW → 0.500W rated ✓

Linear regulators

fig. 2. Group A: DC current gain as a function of collector current; typical values

To obtain the maximum possible current gain using Darlington pair it's important to ensure that the driver transistor has enough collector current to ensure acceptable hFE. When run at very low collector current, most transistors will have a gain that's well below the quoted figure in the datasheet The base-emitter resistor (and [R69](#)) is sized to ensure that the drive transistor has a collector current that's above the minimum. The resistor also helps to ensure a faster turn-off and minimises leakage current.

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